

ExBreath: Explore the Expressive Breath System as Nonverbal Signs towards Semi-unintentional Expression

Yuan-Ling Feng
The Future Laboratory
and Academy of Arts & Design,
Tsinghua University
Beijing, China
fyl20@mails.tsinghua.edu.cn

Mingyue Gao
The Future Laboratory
and Academy of Arts & Design,
Tsinghua University
Beijing, China
gaomy20@mails.tsinghua.edu.cn

Chih-Heng Li
The Future Laboratory
and Academy of Arts & Design,
Tsinghua University
Beijing, China
Chih-Heng22@mails.tsinghua.edu.cn

Zhihao Yao
The Future Laboratory
and Academy of Arts & Design,
Tsinghua University
Beijing, China
yaozh_h@outlook.com

Haipeng Mi*
The Future Laboratory
and Academy of Arts & Design,
Tsinghua University
Beijing, China
mhp@tsinghua.edu.cn

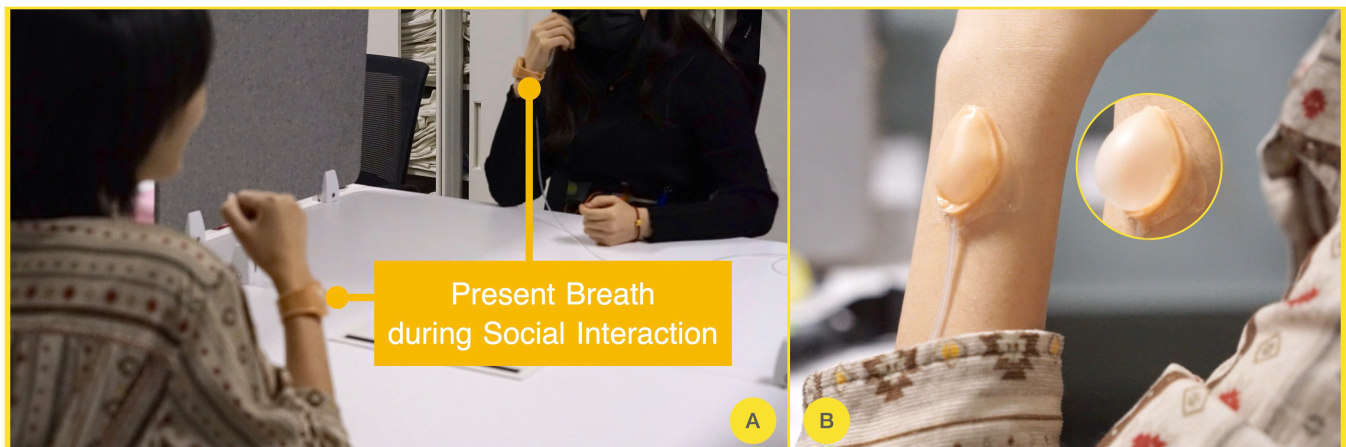


Figure 1: A. Presenting breath in everyday life as an explicit body movement to enhance social interaction. B. Our prototype for amplifying breathing, which can be adhesive on any part of the body.

ABSTRACT

Biosignals, functioning as additional nonverbal signs, have the potential to introduce novel social experiences. The content and presentation forms of biosignals play a crucial role in shaping the final social behavior and user experience. Initiating with the exploration of expressive breath display, we developed ExBreath—an organ-like, inflatable silicon device, which was subsequently utilized in an exploratory study. Based on the findings, breath elicits semi-unintentional expressions, where participants trust that the

signals from ExBreath represent the other's true state, while they deliberately controlled their breath to fulfill various social purposes. Moreover, design opportunities for breath-based interactive systems are revealed, including expanding harmonious communication space, reducing social pressure, and eliciting reflection on cultural norms.

CCS CONCEPTS

• Human-centered computing → Collaborative and social computing devices; Empirical studies in collaborative and social computing; Empirical studies in HCI.

KEYWORDS

Breath, Self-Expression, Breath Sharing, Affective Communication, Social Wearable, Pneumatic Display, Collocated Interaction

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* denotes the corresponding author.

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1 INTRODUCTION

Biosignals are frequently used as social signs in today's HCI research on social interaction, whether in remote or collocated settings. In remote communication, they can fill the absence of non-verbal signs and strengthen the connection between individuals, even sparking curiosity about their partner's state.[15, 17, 22, 26, 28, 35]. In collocated communication, the use of biosignals as social signs in everyday life can bring about interesting social experiences, including enriching emotional expression, prompting reflection on self-expression, and fostering a warm communication space[1, 11–13, 29].

It is noteworthy that biosignal-based interactive systems not only deliver social meanings via the content of biosignals but also that the expression forms of the system significantly influence the social experience. As advocated by Howell et al.[10], more attention should be paid to the material and performative expression forms when designing biosensing technologies. This prompted us to consider expressivity[2] while exploring the social effects of biosignal-based systems.

Choosing breath as a starting point is based on its semi-voluntary feature, which can open an interesting social space that remains largely unexplored. Most of the time, breath is produced without the involvement of conscious thought, delivering unintentional messages continuously, yet it can also be produced intentionally. Once this behavior is made visible, will people fear the exposure of their unconscious states, or will they believe they have everything under control because it can be manipulated?

With such a question in mind, we explored both **1) more expressive ways of presenting breath** and **2) the impact of this system on social interaction**, through a system called **ExBreath** that externalized breath as a bodily movement. After an exploratory study, we found that participants treated ExBreath as a part of self-expression and would control their breath during conversations. Even though breath can be controlled, participants still trusted the information conveyed by the breath and interpreted it in combination with other social signs within the contexts. The findings show that breath brings a controllable yet unintentional form of expression, which we term a semi-unintentional expression.

Our main contributions are:

- An expressive breath display, ExBreath, with soft pneumatic features.
- Findings from the study explore how breath involves and influences collocated social interaction.
- Conclude and reflect on the design opportunities and semi-unintentional expressions expended by ExBreath.

2 RELATED WORK

2.1 Expressive Design on Breath-based Interactive System

The digital presentation of breath can be constructed through various elements and interfaces. As summarized by Prpa et al. [25],

breath can be represented as abstract elements, everyday objects, nature-inspired elements, and human-inspired elements, and then perceived by users through interfaces such as desktop and mobile apps, VR, spatial installation, and everyday objects in both visual, audio, and kinesthetic modality[3, 5, 15, 19, 21, 22, 30, 33].

There were partly works that have been present breath in expressive ways. For example, in the virtual environment, breath can be visualized as a visual effect, a companion pet, a change on the avatar body or environment surrounding the avatar[16], or used as an input signal to control the avatar, such as the expand-contract of a jellyfish avatar[32]. In the real world, Kim et al. share breath through a photo frame embedded with a partner's picture and a pneumatic airbag[15]; Gamboa et al. connect breath to drone movements[6]; an artwork by Maayan Wisstub showcases a living desk through rhythmic and dynamic undulations on its surface, capturing the essence of breathing¹. However, the expression of breath as an extended body part in the real world remains largely unexplored.

2.2 Extending Bodily Communication

Breath as a body movement is difficult to perceive daily because of the limitation of body anatomy, and technologies allow us to amplify it into a body movement that can be perceived. Similarly, some studies have recognized the anatomical limitations of the body and attempted to break through them with technology. One approach is creating new movements on the body by modifying the original body parts or making an additional body extension. For example, researchers tattoo the body surface to visually change the skin[18], or move body parts that would not normally be moved, such as the ears, to express emotions or a newly defined body language[14, 24]. They also create body extensions such as tails[34] or shoulder[8] for abundant expression. The other approach is extending the way we perceive body movements. Beyond the limitation of the distance between physical bodies, we can closely feel others' movements when we are physically at a distance, such as feeling the other's heart beating on the skin[35], even synchronizing body movements with the other kinesthetically[23], to feel others on self's body.

This paper explores the potential of displaying breath as a bodily extension to provide nonverbal signs that are naturally perceived as physical movements. Furthermore, it investigates the experience of not only visually perceiving others' breath but also physically feeling the presence of others' breath on one's skin.

3 EXBREATH: EXTERNALIZING BREATHING AS EXPLICIT BODY MOVEMENTS

3.1 Design

To design a system that smoothly expresses breath through explicit body movement, we delve into understanding how breath naturally influences our body. The Soma Design method[9] provides us with a methodological foundation here. It builds upon somaesthetics theory and takes the body as a non-dualistic and living *soma* rather than a *torso* separated from the mind. Through engaging designers in sensitivity training around the body from a first-person perspective, it aims to improve their aesthetic sensitivity and, ultimately,

¹<https://weisstub.com/all-items/works/mn%E1%B8%97me/>

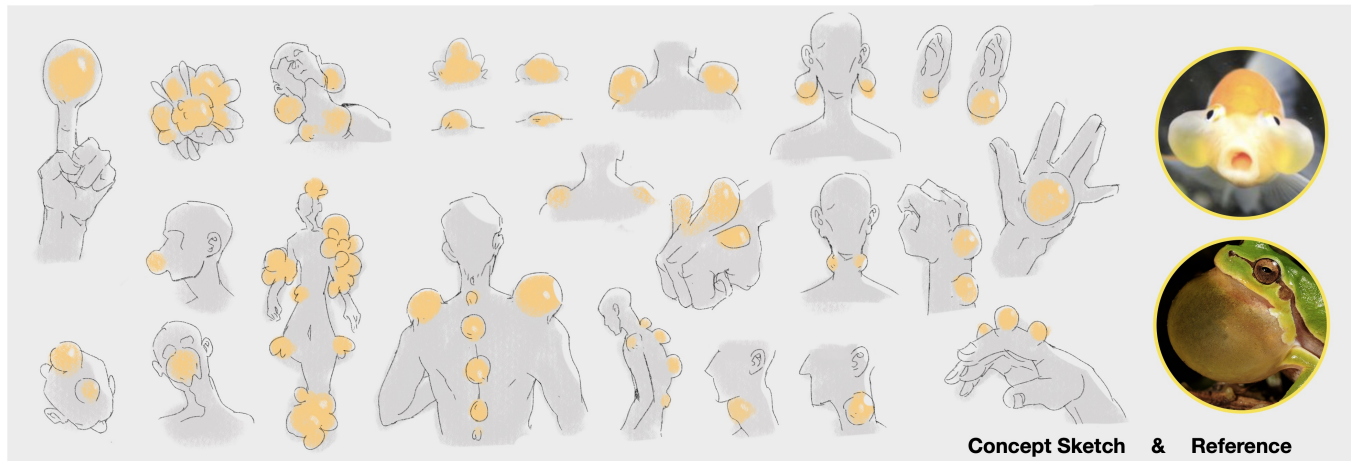


Figure 2: Ideation sketch of breath organs, inspired by fish with bubble eyes and the vocal sac on frogs.

to benefit the design outcome. Therefore, we engage in a series of body exercises, including yoga, contact improvisation, and kung fu. Through first-person experiences and interviews with body experts, we discover that the subtle sensations and metaphors brought about by the dynamic of breath are much richer than we initially imagined before experiencing them. The fundamental understanding of breath involves *the continuous flow throughout the body and the smooth expansion-contraction of the entire body*.

Therefore, when translating the dynamic of breath into a distinct non-verbal sign, we introduced the concept of a "breathing organ", mapping the smooth expansion-contraction of the entire body onto an undulating external organ. The conceptualization of this organ was also inspired by the structural features of animal bodies, as illustrated in figure 2, where textures resembling frog vocal sacs and fish swim bladders closely approximate the envisioned texture of our imagined breathing organ. Building on this idea, we further envisioned the imagery of breathing organs growing from different parts of the body.

Following the concept, we start to explore the materials that can support the aesthetic qualities. This breathing organ needs to exhibit a soft and touchable skin texture, and the inflation process, synchronized with body movements, should ideally respond flexibly to subtle changes while maintaining tranquility, vitality, and an organic feel throughout the movement. After trying various materials and fabrication methods, including flexible 3D printing, latex balloon skin, and off-the-shelf silicone films, we ultimately decided to use EcoFlex silicone for the entire air bubble. This method allows the creation of a very thin and rupture-resistant bubble surface. After inflation, the surface can visibly stretch and thin, presenting a form close to that of a bubble with a more organic feel (figure3, left). Simultaneously, this choice provides flexibility for customizing organ shapes and allows the use of pigments for coloring. The specific fabrication process can be summarized in three steps: making molds for three components separately, curing silicone, and bonding the three components together. The three components are shown in Figure 3.a, from left to right: the stretched surface, the middle layer with a hole in the center and a half-channel going through, and the

bottom layer with the other half of the channel. After the silicone is cured, we remove the bottom and middle layers from the molds first, then place an air tube between them and apply freshly mixed liquid silicone to bond the two parts together. Then, we apply a ring of silicone along the top and side edges of the middle layer to bond the surface. The customized bubbles can be attached to different body parts by adhesive, moreover, prosthetic makeup techniques could smooth the visual effect of the adhesion area and make the bubble a vivid extended body part.

The selection of the inflation module also underwent comparative screening because it affects the aesthetic experience of bladder movement. In the end, we chose a micropump based on the ultrasonic vibration of ceramics because it can inflate and deflate very quietly and sensitively

3.2 Technical Realization

Each set of ExBreath consists of three parts which are drawn in figure 4. Breath data is detected by a flexible bend sensor mounted on a sports belt (fig.5.a), and then transferred to the PC where mapping the raw data to actuation commands. The program on PC is developed on Touch Designer (fig. 5.c), corresponding with breath data to control the inflating/deflating speed of actuation module and switch it between inflating, deflating, and closing. The actuation module consists of an Arduino, two solenoid valves (SMC S070C-6AG-32), and a micropump (MURATA MZB3004T04) (fig.5.b).

4 EXPLORATORY STUDY

4.1 Setups

We conducted a pilot study to assess our system's impact on social interactions, with participants wearing the device during conversations, followed by interviews using the Explication Interview method [20]. This method helped us explore experiences influenced by breath sharing, particularly focusing on behaviors, thoughts, and awareness of others' breath. For natural interaction and credible data collection without audio or video recording, we included a researcher disguised as a participant. The study, targeting interactions with strangers, was held in a private room, as shown in figure

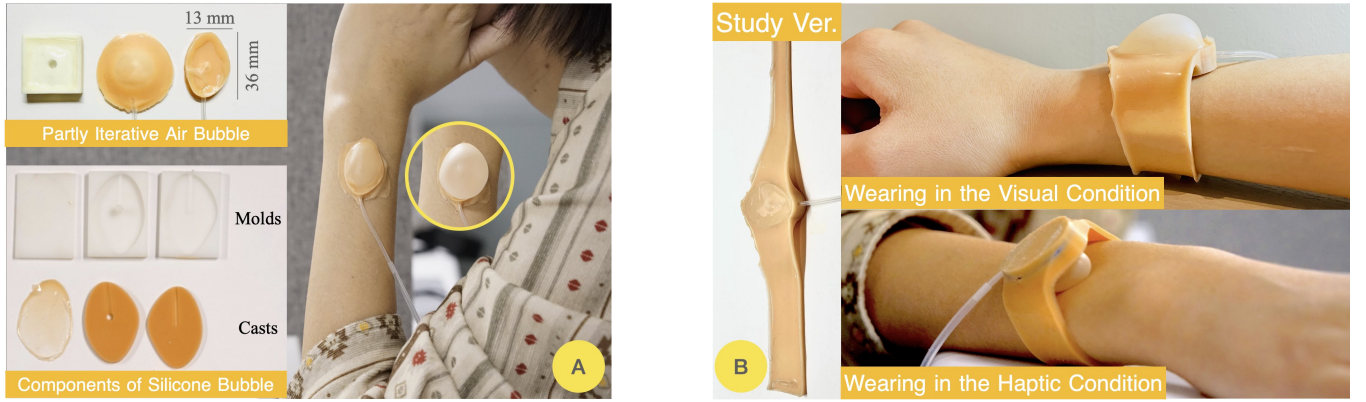


Figure 3: A: iteration and components of silicon bubbles. B: Modified bubbles for studying the influence on social interaction in visual and haptic conditions.

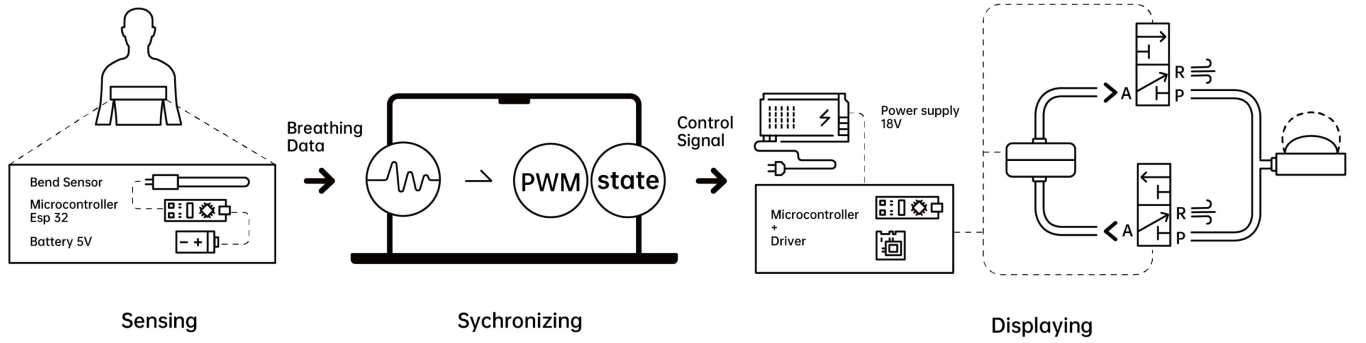


Figure 4: System structure of ExBreath, consists of sensing, synchronizing, and displaying parts. The dotted line between electronic modules is the wire, and the hollow line is the air tube.

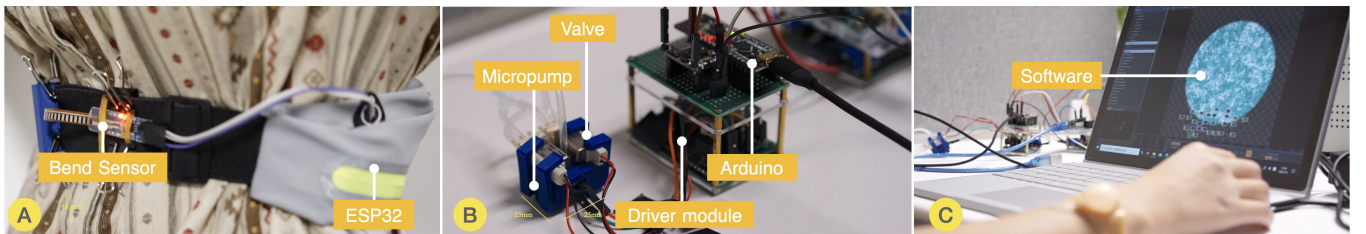


Figure 5: A. Breath Sensor B. Actuation module for controlling silicone bubble C. Software for Synchronizing

6, where participants conversed across a table while equipped with our devices.

To enhance participant interaction involving sequential seeing and touching, we crafted a unique design of ExBreath, which has an easily wearable bubble with a wrist strap, depicted in figure 3.b. This design is adapted for shifting more easily between visual and haptic experiences rather than directly adhering to the skin. We also integrated a hard ABS slice at the base of the bubble, enabling it to inflate on only one side. This feature allows participants to wear the bubble in two different orientations: either to visually showcase their breath or to tactfully sense another's, as illustrated in figure 6, right.

4.2 Procedures

Twelve participants (6f, 6m, age $M=23$, $SD=1.8$, and coded as 1-12F/M) are invited through online recruitment with half starting from the visual condition, and the other starting from the haptic condition.

Upon a participant's arrival, the experimenter guides them to a seat, instructing them to wear a breath-sharing device and engage in a conversation with another participant (the interlocutor). After being seated and assisted with donning the breath sensor and display, the interlocutor is similarly equipped. The experimenter then prompts a casual conversation about introductions and hobbies

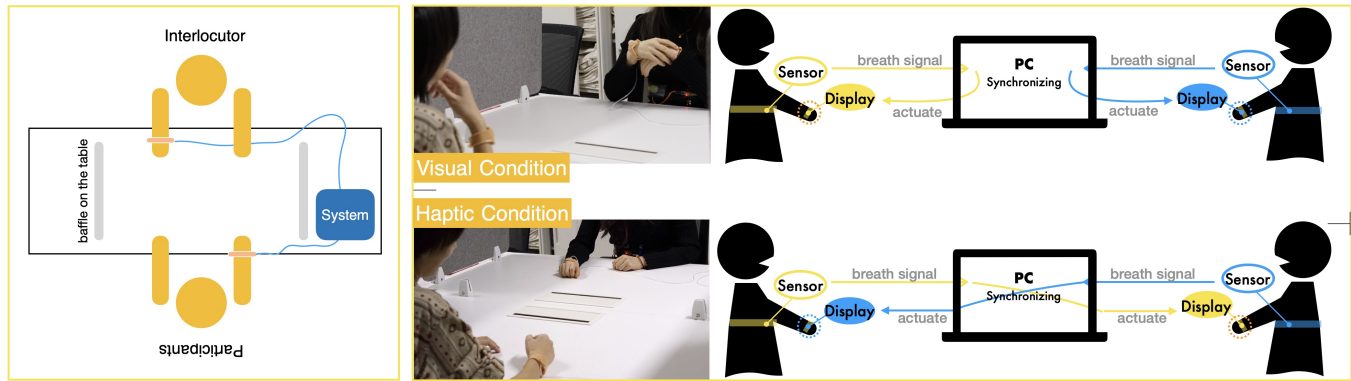


Figure 6: Left: Study setup in top view. Right: Two conditions to share breath: in the visual condition (top), participants can see the other’s breath on the other’s wrist; in the haptic condition (bottom), participants can perceive the other person’s breathing state through the display mounted on their own wrists.

before leaving the room. The conversation lasts 3 minutes per condition, based on pilot study findings that most feedback occurs in the first minute, and longer conversations challenge recall accuracy. After 3 minutes, the experimenter re-enters to assist with device adjustments and condition switching. Post-discussion, the participant is interviewed individually for 40 to 60 minutes, followed by a joint interview with the interlocutor to observe and discuss the participant’s behavior.

4.3 Findings

Interview data were transcribed and coded in MAXQDA, and findings from it focus on the expressivity of the interaction with the prototype, the difference between two presenting conditions, and the influence on the interpretation and expression of participants.

4.3.1 Expressivity of the Interaction. Overall, participants naturally perceive the dynamics of ExBreath, seamlessly integrating it into their communication as an extension of their bodies. Participant 8M noted, “When I was wearing it, I felt it stopped when I held my breath, and it inflated when I inhaled, so I knew it reflects my breath.” As conversations progress, its presence often becomes less noticeable, with participants feeling it as “integrated with me, I usually forgot it”. Most participants found the skin-colored bubble of ExBreath to have an organic texture, describing it as “alive,” “biological,” or similar to “alveolus” or “air bubbles beside fish gills,” with one seeing it as an independent organism. A few, however, simply saw it as “just a watch.” Regardless, all agreed it stood out distinctly from typical screen-based digital devices.

4.3.2 Difference between Visual and Haptic Condition. Participants primarily focused on conversation, but sudden changes in the device’s speed and amplitude captured their attention. As 6F noted, “I could feel at irregular changes [...] the inflation going stronger or disappeared.” This awareness made them more attuned to subtle changes in the other’s state.

The differences in perception across conditions centered on attention, intensity, and intuition. In the haptic condition, breath was felt as a continuous stimulus, demanding less attention than in the visual condition, where participants had to actively glance

at the device. Some preferred the active engagement of the visual condition, while others favored the haptic condition for its passive, ongoing awareness.

The haptic stimuli, pressing directly against the skin, felt more intense than visually observing undulations from a distance. This tactile sensation brought participants, like 5M, closer to the interlocutor’s breath. In terms of intuition, participants found both visual and haptic displays intuitive: haptically, they directly felt the breath; visually, they quickly understood breath states by observing the silicon cell’s inflation. As 12F mentioned, “I knew (the breath state) just take a look [...] if it (breath) is displayed as a number, I need to think the relationship between the number and the body state.”

In both conditions, breath involved conversation successfully, moreover, altered participants’ understanding of the current social situation, and changed their expression response to the situation, including speaking, action, and breath.

4.3.3 Interpreting Expression with Breath. ExBreath enhanced participants’ confidence in interpreting others, as they trusted the device’s ability to accurately capture genuine internal states over their own judgments. For example, 4F mentioned, “I remained in doubt of what I observe on her... I trust the information from the device is true.” and 6F noted, “you can cover up emotion, but it’s hard to adjust breath immediately.”

Participants linked breath to cognitive and emotional information. Cognitively, breath changes indicated engagement in conversation, as 2M and 3F observed. Emotionally, breath patterns were seen as reflecting emotions, with frequent associations between breath and nervous or calm states. 6F observed, “Strong inflation indicated nervousness,” while 12F contemplated, “smaller amplitudes meant the interlocutor was calmer.” Besides nervousness, upheaval in breath was also interpreted as urgency or resentment.

4.3.4 Change on the Self-expression. ExBreath affected participants’ self-expression, both deliberately and inadvertently, altering their speech, actions, and breathing. Participants often consciously modified their speech and actions; for instance, 11F spoke more cautiously when aware of possibly offending the interlocutor. In some

cases, these changes were involuntary. For example, 4F, with strong social anxiety, felt relaxed by the device's calm undulations, which facilitated easier conversation. This effect, particularly evident in the haptic condition, significantly influenced social interactions and behaviors, as 4F felt reassured by the tangible sensation of calm breath.

Participants employed various strategies for breath control, including inhibiting, alternative exposure, and honest exposure. **Inhibiting** involved calming breath to comfort themselves or others and hide nervousness. **Alternative exposure** was about intentionally making breath more noticeable, aligning it with emotional expression for sincerity. 2M commented, "*if my words surprise someone, I want the device to change, to expand.*" This method also conveyed unspeakable information. **Honest exposure** meant presenting unaltered breath, seen as a way to foster honesty and mutual understanding.

5 DISCUSSION AND REFLECTION

5.1 Breath bring Semi-unintentional Expression

From the findings, it is clear that breathing impacts both interpreting others and expressing oneself. People combine breathing to gain a new understanding of each other's state, and they trust the information conveyed by breathing and control it for particular social purposes. Although breathing can be controlled, participants continue to trust that the other person's breathing conveys their true state. This breath-interfered expression is close to unintentional expression in its truthfulness but has the controllability of intentional expression, which we call semi-unintentional expression. It is characterized by 1. continuously reacting to personal states when an individual is unaware and 2. being easily and instantly controlled when an individual is aware.

5.2 Design Opportunities Extended by ExBreath

5.2.1 Expanding Communication Space. Breath-based interactive systems expand the space for interpretation and expression, potentially fostering more harmonious conversations by providing a subtle means to convey messages. From the perspective of expression, individuals can use it to make others aware of their true thoughts while avoiding disputes. On the interpretation front, semi-unintentional expression blurs the interpretive space, as others cannot confirm whether the expression is intentional or unintentional. Expanding the space of expression and interpretation provides people with more room to maneuver in cooperation toward the common goal of smooth social interaction. The semi-unintentional expression brought by breath increases resilience in social interaction.

5.2.2 Releasing Social Pressure. The study noted a relaxing effect from ExBreath, suggesting its use in reducing social pressure and social anxiety. Participant 4F relaxed due to two factors: first, ExBreath indicated the calmness of her conversation partner, which is significant for those with social anxiety who fear negative judgments[31]. Second, the social touch aspect of the device, made of silicon, likely subconsciously influenced her, as social touch often relieves stress

and improves perception[27]. These findings suggest that breath-sharing systems with elements of social touch could be beneficial in situations like therapy, public speaking, and meeting strangers.

5.2.3 Reevaluating Cultural Norms and Social Behavior. Controlling breath is a form of **impression management**[7], closely tied to how individuals believe they should present themselves in specific situations. People adjust their behavior based on their understanding of cultural norms to fit into particular contexts. In our daily lives, our bodily actions have become habitual, making it challenging to distinguish between innate and culturally conditioned behaviors. However, as breathing becomes a new social sign, we need to regain control over our bodies. This allows researchers to observe, and even prompts users to reflect on, which behaviors are influenced by cultural norms and which represent their most authentic expressions. This contributes to encouraging behavioral changes, as misunderstandings of cultural norms can sometimes lead to excessive control of expression, becoming a barrier to effective communication[4, 36].

6 LIMITATION AND FUTURE WORKS

Although we found a significant influence of sharing breath in co-located communication through this study, some limits can be improved in the future.

Firstly, the daily chat scenario for the user study provided a peaceful and usual ambient for conversation. While participants described smooth, flat conversations with stable, calm emotions, this scenario limited emotional abundance, avoiding conflict or embarrassment. As excitement or anger did not emerge, further exploration is needed on how sharing breath affects communication under negative or unstable emotional states.

Secondly, the exploratory study modified the silicone bubble, making it more wearable but less reminiscent of an organ, which deviated slightly from the original design intention. Perhaps alternative approaches, such as speculative design, could be employed to investigate the social implications and perceptions of a breath-based organ-like artifact.

Thirdly, the prototype can be improved by enhancing the breath sensor's sensitivity to detect subtle changes and integrating the display into a fully wireless device, enabling long-term field studies. As people learn to socially control expressions over time, we hypothesize that externalizing breath will increase their breath control skills and potentially alter their attitudes toward it. Breath expressions may become less believable with prolonged use, further motivating future long-term studies.

7 CONCLUSION

This paper introduces a design case named ExBreath, which explores the expressive design of a social breath system. Through an exploratory study, it uncovers the impact of ExBreath on both interpretation and expression in social situations. The findings underscore that breath-based interactive systems create a unique social experience rooted in semi-unintentional expression, thereby expanding design opportunities. These opportunities include fostering more harmonious conversations, reducing social pressure, and reflecting cultural norms.

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